RACKSEPHNOX

Global Cryptocurrency | Fintech | Gaming Empire

Framework Documentation
Technical Reference

Every System | Every Module | Every Workflow

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1. Platform Overview

Racksephnox is a unified digital wealth platform combining cryptocurrency trading, investment products, gaming, loans, and referral systems into a single authenticated experience. Built on Laravel 13 with a mobile-first philosophy and deployed globally via Cloudflare Tunnel.

Core Design Principles

- Unified identity - one account for every module
- Transparent operations - full audit trail on every action
- Mobile-first - designed for emerging markets
- Modular architecture - each framework independent
- Golden theme - gold, black, ivory palette

Platform Statistics

Frameworks	15+ distinct systems
Database Tables	60+ tables
Controllers	40+ across web and API
Services	30+ business logic classes
Models	50+ Eloquent models
Views	80+ blade templates
API Endpoints	60+ REST endpoints
Lines of Code	50,000+

2. Technology Stack

Language	PHP 8.4
Framework	Laravel 13
Database	SQLite (dev) / PostgreSQL (production)
Frontend Build	Vite 5
CSS Framework	TailwindCSS 3
JS Framework	Alpine.js + Vanilla
Charts	KLineChart (candlesticks)
Icons	Font Awesome 6 + Emoji
Authentication	Laravel Breeze + Sanctum
Deployment	Cloudflare Tunnel
CI/CD	GitHub Actions
Container	Docker multi-stage
Cache	File (dev) / Redis (production)
Queue	Database / Redis
Mail	Log (dev) / SMTP (production)

3. Authentication Framework

Purpose

Handles user registration, login, session management, and access control. Built on Laravel Breeze with custom enhancements for admin detection.

Database Tables

users	Core user records
password_reset_tokens	Password reset flow
personal_access_tokens	Sanctum API tokens
sessions	Active web sessions

Controllers

- AuthenticatedSessionController - login/logout
- RegisteredUserController - new user creation
- PasswordResetLinkController - forgot password
- NewPasswordController - reset token validation
- EmailVerificationPromptController - verify email

Workflow - Registration

- User submits name, email, password
- Validation ensures unique email and strong password
- User record created with auto-verified email
- Wallet + TradingAccount + TradingProfile created via boot()
- Referral code auto-generated
- User redirected to dashboard

Workflow - Login

- User submits email + password
- Rate limiter prevents brute force (5 attempts)
- On success, session regenerated
- Admin users redirected to /admin
- Regular users redirected to /dashboard

Security Features

- Bcrypt hashing with 12 rounds
- CSRF tokens on every form

- Rate limiting on auth routes
- Email verification option
- Two-factor authentication schema ready

4. Dashboard Framework

Purpose

Central hub after login. Aggregates data from every subsystem into a unified overview: wealth, banking, investments, loans, referrals, gaming.

Sections

- Wealth Pillars - balance, invested, profit, active holdings
- Quick Navigation - 11 portals to every module
- Lottery Jackpot - live progressive pool
- Banking Center - deposit and withdrawal CTAs
- Loans Widget - credit score, upcoming repayments
- Referral Widget - count, bonus earned, share link
- Trading Widget - balance, BTC price
- Machine CTA - RX Series promo
- Charts - 30-day profit, weekly activity, portfolio
- Recent Transactions - last 10 entries

Data Aggregation

DashboardController::index() queries 12+ tables and caches heavy aggregates (BTC price, credit score, jackpot total) for 30-300 seconds.

Real-Time Updates

- Unread notifications: every 30s
- BTC price: every 10s
- Lottery jackpot: every 15s

5. Banking Center (Deposits & Withdrawals)

Purpose

Handles all money movement in and out of the platform. Users deposit via M-Pesa Paybill 200999 and withdraw to their mobile or bank account.

Database Tables

deposit_requests	User deposit submissions
withdrawal_requests	User withdrawal requests
mpesa_transactions	Raw M-Pesa transaction log
user_bank_accounts	Saved bank accounts for withdrawals
wallets	KES wallet per user

Deposit Flow - Complete

- User opens /deposit
- Sees Paybill 200999 and Account 0333060036213
- Sends money via M-Pesa app
- Receives SMS with transaction code (e.g. SFE4A8B2C1)
- Enters code + amount on the deposit form
- System validates code format (10 chars, starts with letter)
- System checks code is not duplicated
- System attempts auto-match against uploaded statements
- Deposit stored with status = pending_verification
- User sees 'Awaiting Verification' page

Admin Verification Flow

- Admin opens /admin/deposits
- Sees pending list with code, amount, phone, user
- Option A: Manually verify against M-Pesa app
- Option B: Upload M-Pesa statement CSV - auto-matches
- Admin clicks Verify - wallet credited - transaction logged
- User notified via in-app notification

Withdrawal Flow

- User requests withdrawal with amount + account
- System checks balance, caps, and KYC status
- Wallet debited immediately (locked_balance)

- Admin approves and sends via M-Pesa B2C
- Wallet unlocked - transaction completed

6. M-Pesa Integration & Verification

Paybill Details

Paybill Number	200999
Account Number	0333060036213
Transaction Code Format	10 chars, starts with letter
Example Code	SFE4A8B2C1

Verification Framework

Two modes work together: manual verification and automated statement matching.

Manual Verification

- Admin opens M-Pesa app on phone
- Confirms transaction code exists in statement
- Clicks Verify in admin panel
- Wallet credited instantly

Automated Statement Matching

- Admin exports M-Pesa statement as CSV
- Uploads via admin panel
- System parses every transaction row
- Auto-matches against pending deposits
- Confidence scores: 100 = exact, 90 = code only, 60 = fuzzy
- Admin reviews and approves matched rows

Services

- DepositVerificationService - core verification logic
- MpesaStatementParser - CSV parsing

Security Guardrails

- Code format validation (regex)
- Duplicate code detection
- Amount range validation (KES 10 to 1,000,000)
- Phone number format check
- Admin approval required for all credits

7. Cryptocurrency Trading Framework

Purpose

Full-featured crypto trading console with real-time candlestick charts, order book, buy/sell engine, and portfolio management. Currently supports BTCUSDT with architecture ready for ETH, BNB, and others.

Database Tables

trading_pairs	Available pairs (BTCUSDT, ETHUSDT...)
trade_orders	User buy/sell orders
trading_candles	OHLCV candle data
trading_accounts	Per-user trading balance
btc_price_history	Historical BTC price data

Chart Rendering

- Library: KLineChart (MIT, ~40 KB gzipped)
- Delivered via Vite (bundled, no CDN)
- 6 timeframes: 1m, 5m, 15m, 1h, 4h, 1d
- 7 indicators: MA, EMA, BOLL, MACD, RSI, KDJ, OBV
- ResizeObserver for responsive canvas
- requestAnimationFrame throttling for 60 FPS
- Preloads adjacent intervals during idle time

Order Flow

- User selects side (buy or sell)
- Enters amount and order type (market/limit/stop)
- Optional take-profit and stop-loss
- System validates against balance
- Market orders execute immediately at current price
- Limit orders placed in the book for matching

Services

- TradingEngine - market price, order book, order placement
- ChartService - Binance sync + synthetic fallback
- TradingBonusService - signup bonus + volume tracking

Real-Time Updates

- Candles refresh every 5 seconds

- Order book refreshes every 3 seconds
- Auto-reconnect with exponential backoff
- Pauses when browser tab in background

8. Forex Trading Framework

Purpose

Extension of the crypto trading engine for traditional forex pairs. Shares the same order book, engine, and chart architecture, with forex-specific configurations.

Planned Pairs

- EUR/USD - most liquid globally
- GBP/USD - major pair
- USD/JPY - Asian session driver
- USD/KES - Kenya-focused

Implementation Notes

- Same TradeOrder table with pair_id differentiation
- Same ChartService - configure forex data source
- Leverage tiers configured per pair
- Spread applied to entry/exit prices

Regulatory Considerations

- Forex is regulated in most jurisdictions
- KYC required above certain volumes
- Compliance officer role assigned

9. Investment Plans Framework

Purpose

Traditional investment plans where users lock KES for a duration and earn daily interest. Interest is credited automatically every 24 hours.

Database Tables

investment_plans	Plan definitions
investments	User investments

Default Plans

Starter	KES 1K-50K | 1.5%/day | 30 days
Growth	KES 10K-250K | 2.25%/day | 30 days
Premium	KES 50K-1M | 3%/day | 45 days
Divine	KES 100K-5M | 6.28%/day | 14 days (88% ROI)

Workflow

- User selects plan and enters amount
- System validates against plan min/max
- Wallet debited immediately
- Investment record created with projected totals
- Daily profit accrues via scheduled job
- User can view progress and remaining days
- At maturity, user can withdraw total returns

Services

- InterestCalculator - daily/total profit math
- InvestmentService - creation, accrual, withdrawal
- InvestmentManager - plan stats and user holdings
- UnifiedInvestmentService - combines with machines

10. RX Machine Series

Purpose

Premium investment product with 7 tiers (RX0 through RX6) and 3 VIP levels per tier. Higher tiers offer higher returns and higher minimums.

Database Tables

machines	Machine tier definitions
machine_vips	3 VIP levels per machine
machine_investments	User holdings

The 7 Machines

RX0 Origin	KES 1K-25K | 25% growth
RX1 Aurora	KES 2K-50K | 30% growth
RX2 Nova	KES 5K-100K | 38% growth
RX3 Prism	KES 10K-250K | 45% growth
RX4 Eclipse	KES 25K-500K | 55% growth
RX5 Quantum	KES 50K-1M | 70% growth
RX6 Infinity	KES 100K-2M | 88% growth

VIP Tiers per Machine

- VIP 1: Baseline growth rate
- VIP 2: Lambda-multiplied growth (x1.272)
- VIP 3: Phi-multiplied growth (x1.618)

Golden Constants

- phi = 1.61803398875 - divine proportion
- lambda = 1.27201964951 - sqrt(phi)
- pi = 3.14159265359 - circular constant
- e = 2.71828182846 - natural growth

Cycle

- 14-day investment cycle
- Daily profit accrual
- Early withdrawal penalty: 20%
- Referral bonus: 5%

11. BTC Investment Machines

Purpose

Bitcoin-specific investment machines that track BTC price appreciation and combine with trading profits. Shares the Machine architecture but uses BTC-based returns instead of fixed percentages.

Mechanism

- User invests KES equivalent of BTC
- Machine tracks BTC price in real-time
- Profit = BTC price appreciation + machine bonus
- 14-day cycle, same as RX Series
- Auto-compound option available

Integration

- Uses trading_candles for BTC price data
- Syncs with TradingEngine::getMarketPrice()
- Shares machine_investments table with tier flag

12. Lottery & Casino Framework

Purpose

Full gaming ecosystem with provably-fair spins, progressive jackpots, achievements, missions, streaks, tournaments, and guilds. Designed for engagement and retention.

Database Tables (18)

lottery_games	Game definitions
lottery_symbols	Reel symbols
lottery_payouts	Payout matrix
lottery_spins	Every spin record
lottery_jackpot_pools	4-tier progressive pools
lottery_jackpot_wins	Jackpot win history
lottery_achievements	Milestones
lottery_user_achievements	Earned achievements
lottery_daily_streaks	Streak tracking
lottery_missions	Daily quests
lottery_user_missions	Quest progress
lottery_tournaments	Competitions
lottery_tournament_entries	Scores
lottery_guilds	Player collectives
lottery_guild_members	Membership
lottery_bonus_wheels	Wheel config
lottery_bonus_wheel_spins	Wheel history
lottery_fair_seeds	Provably-fair chain

4-Tier Jackpot System

Bronze	528 Hz | seed 10K | ceiling 500K
Silver	639 Hz | seed 100K | ceiling 5M
Gold	741 Hz | seed 1M | ceiling 50M
Cosmic	963 Hz | seed 10M | ceiling 100M

Provably Fair

- Each spin produces client_seed + server_seed + nonce
- HMAC-SHA256 produces verifiable hash
- User can rotate seed chain anytime
- Verification endpoint allows historical audit

Services

- RngService - provably-fair random generation
- LotteryService - spin orchestrator
- JackpotPoolService - 4-tier pool management
- AchievementService - milestone detection
- StreakService - daily streak rewards
- MissionService - quest generation
- BonusWheelService - weighted wheel
- TournamentService - scoring and prizes
- GuildService - collective management

Responsible Gaming

- Daily/weekly/monthly loss caps
- Cool-down periods
- Self-exclusion (24h to 1 year)
- Session timeout at 60 min
- Reality checks every 30 min

13. Loans Services Framework

Purpose

Lending platform with credit scoring, tiered products, amortization scheduling, and default detection. Users build credit through on-time repayment.

Database Tables

loan_products	Product definitions
loans	Active and historical loans
loan_repayments	Amortization schedule
loan_guarantors	Guarantor records
loan_collaterals	Collateral registry
loan_credit_scores	User credit scores

Default Products

Divine Micro	KES 500-20K | 8% flat | 7-30 days
Golden Personal	KES 5K-200K | 15% reducing | 30-180 days
Phi Business	KES 50K-2M | 12% reducing | 90-730 days
Infinity Mortgage	KES 500K-20M | 9.5% | 1-10 years

Credit Scoring

- Base score: 500
- +200 for full repayment ratio
- +5 per on-time payment
- -10 per late payment
- -150 per default
- Range: 300 to 850

Workflow

- User applies - amount, duration, frequency
- System checks credit score and exposure
- Admin approves or rejects
- Admin disburses - wallet credited
- Amortization schedule generated
- User repays - schedule updates
- Credit score recalculated after each payment

14. Referral System

Purpose

- Growth engine - users earn bonuses for inviting others

Mechanism

- Every user gets a unique referral code
- Share link: /register?ref=CODE
- Referrer earns 5% of referee's first investment
- Bonus credited directly to wallet
- Referral count and total earned shown on dashboard

Database Fields

- users.referral_code - unique per user
- users.referred_by - parent referrer ID
- transactions where type = 'referral_bonus'

15. Notifications Framework

Purpose

Keeps users informed about transactions, wins, loans, and events via in-app, email, and SMS channels.

Channels

- In-app (database channel)
- Email (SMTP)
- SMS (future - Africa's Talking integration)

Events That Trigger Notifications

- Deposit confirmed
- Withdrawal processed
- Big win (lottery)
- Jackpot won
- Investment matured
- Loan approved/rejected
- Referral bonus earned
- KYC approved/rejected

Preferences

- User can opt out per channel
- Preferences stored in users.notification_preferences JSON
- Unread count cached and polled every 30s

16. KYC & Verification

Purpose

Know Your Customer - required for regulatory compliance and to prevent fraud. Users submit identity documents for verification.

KYC Levels

Basic	Email + phone verified
Standard	National ID + selfie uploaded
Enhanced	Proof of address + bank statement
VIP	Full KYC + source of funds

Database Tables

kyc_documents	Uploaded documents
users.kyc_status	Current status
users.kyc_level	Verification tier

Workflow

- User uploads ID document
- Admin reviews in /admin/kyc
- Approve - kyc_status = verified
- Reject - user notified with reason
- Higher levels unlock larger limits

17. User Profile & Settings

Purpose

Central page for user account management - personal info, password, notification preferences, bank accounts, and security settings.

Sections

- Personal information (name, email, phone)
- Password update
- Two-factor authentication
- Notification preferences
- Bank account management
- Session management
- Account deletion

Security

- Password confirmation required for sensitive changes
- Session invalidation on password change
- Email notification on every major change

18. Admin Console

Purpose

Back-office for managing every aspect of the platform. Nine department admins each have dedicated sections tailored to their responsibilities.

Admin Sections

- Dashboard - platform-wide stats and activity
- Users - list, edit, suspend, verify
- Deposits - verify, reject, upload statements
- Withdrawals - approve, process, reject
- KYC - review documents
- Investments - manage plans
- Machines - manage RX tiers
- Trading - monitor order book
- Lottery - games, symbols, jackpots, analytics
- Loans - approve, reject, disburse
- Reports - export CSV/PDF
- Settings - platform configuration

Access Control

- Admin middleware on every /admin route
- is_admin flag in users table
- Separate layout (admin.layouts.app)

19. Deployment & CI/CD

CI/CD Pipeline

- GitHub Actions runs on every push to spiral branch
- Workflow: Laravel CI - runs tests
- Workflow: Docker Build - builds multi-arch image
- Image pushed to GitHub Container Registry (GHCR)
- Both run in parallel, cancel outdated runs

Docker

- Multi-stage build - vendor, frontend, runtime
- Alpine base - small image size
- PHP 8.4 FPM with all extensions
- OPcache enabled for production
- Non-root user (www-data)
- Health check endpoint

Cloudflare Tunnel

- Public HTTPS URL without public IP
- Named tunnel for racksephnox.us.ci
- Auto-reconnect on network changes
- DDoS protection and WAF included
- Free SSL on edge

20. Security Architecture

Application Security

- CSRF protection on all forms
- SQL injection prevention via Eloquent
- XSS prevention via Blade escaping
- Password hashing with bcrypt 12 rounds
- Rate limiting on auth and sensitive routes
- Two-factor authentication schema ready
- Session encryption and regeneration

API Security

- Sanctum tokens for API authentication
- Token expiry and rotation
- Rate limiting per user and per IP
- CORS configured for known origins

Data Protection

- Sensitive fields encrypted at rest
- Database backups scheduled
- Audit log for every admin action
- GDPR-compliant data export/deletion

Financial Security

- All balance changes within transactions
- Row locking on wallet operations
- Duplicate transaction prevention
- Referral abuse detection
- Withdrawal approval required above threshold

Infrastructure Security

- Cloudflare DDoS protection
- SSL/TLS on all traffic
- Non-root container user
- Firewall rules on production server

The Empire Documented

Twenty frameworks. One unified vision. Racksephnox is not just software - it is an operating system for digital wealth in emerging markets.

Racksephnox

Infinite Spiral of Creation | 888 Hz | Golden Ratio Phi